

Your grade: **96.66%**

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Next item →

1. If searching among a large number of hyperparameters, you should try values in a grid rather than random values, so that you can carry out the search more systematically and not rely on chance. True or False?

1 / 1 point

- ☐ True
☒ False

✔ Nice work

2. Every hyperparameter, if set poorly, can have a huge negative impact on training, and so all hyperparameters are about equally important to tune well. True or False?

1 / 1 point

- ☒ False
☐ True

✔ Nice work

Yes. We've seen in the lecture that some hyperparameters, such as the learning rate, are more critical than others.

3. During hyperparameter search, whether you try to babysit one model ("Panda" strategy) or train a lot of models in parallel ("Caviar") is largely determined by:

1 / 1 point

- ☐ The presence of local minima (and saddle points) in your neural network
☒ The amount of computational power you can access
☐ The number of hyperparameters you have to tune
☐ Whether you use batch or mini-batch optimization

✔ Nice work

4. Knowing that the hyperparameter α should be in the range of 0.00001 and 1.0, which of the following is the recommended way to sample a value for α ?

1 / 1 point

☐

```
r = np.random.rand()
alpha = 0.00001 + r*0.99999
```

☒

```
r = -5*np.random.rand()
alpha = 10**r
```

☐

```
r = np.random.rand()
alpha = 10**r
```

☐

```
r = -4*np.random.rand()
alpha = 10**r
```

✔ Nice work

Yes. This will generate a random value between 10^{-5} and 10^0 chosen randomly in a logarithmic scale.

5. Once good values of hyperparameters have been found, those values should be changed if new data is added or a change in computational power occurs. True/False?

1 / 1 point

☐

False

☒

True

✔ Nice work

Correct. The choice of some hyperparameters such as the batch size depends on conditions such as hardware and quantity of data.

6. When using batch normalization it is OK to drop the parameter $W^{[l]}$ from the forward propagation since it will be subtracted out when we compute $\tilde{z}^{[l]} = \gamma z_{\text{normalize}}^{[l]} + \beta^{[l]}$. True/False?

1 / 1 point

- ☒ False
- ☐ True

✔ Nice work

Correct. The parameter $W^{[l]}$ doesn't get subtracted during the batch normalization process, although it gets re-scaled.

7. Which of the following are true about batch normalization?

1 / 1 point

- ☐ The parameters β and γ of batch normalization can't be trained using Adam or RMS prop.
- ☐ The parameter ϵ in the batch normalization formula is used to accelerate the convergence of the model.
- ☐ There is a global value of γ and β that is used for all the hidden layers where batch normalization is used.
- ☒ One intuition behind why batch normalization works is that it helps reduce the internal covariance.

✔ Nice work

Yes. Internal covariance is a name to express that there has been a change in the distribution of the activations. Since after each iteration of gradient descent the parameters of a layer change, we might think that the activations suffer from covariance shift.

8. Which of the following is true about batch normalization?

1 / 1 point

- ☐ The optimal values to use for γ and β are $\gamma = \sqrt{\sigma^2 + \epsilon}$ and $\beta = \mu$.
- ☐ $z_{\text{norm}}^{(i)} = \frac{z^{(i)} - \mu}{\sqrt{\sigma^2}}$.
- ☒ The parameters $\gamma^{[l]}$ and $\beta^{[l]}$ set the variance and mean of $\tilde{z}^{[l]}$.
- ☐ The parameters $\gamma^{[l]}$ and $\beta^{[l]}$ can be learned only using plain gradient descent.

✔ Nice work

Correct. When applying the linear transformation $\tilde{z}^{(i)} = \beta^{[l]} z_{\text{norm}}^{(i)} + \gamma^{[l]}$ we set the variance and mean of $\tilde{z}^{[l]}$.

☐ The parameters $\gamma^{[l]}$ and $\beta^{[l]}$ can be learned only using plain gradient descent.

✔ Nice work

Correct. When applying the linear transformation $\tilde{z}^{(i)} = \beta^{[l]} z_{norm}^{(i)} + \gamma^{[l]}$ we set the variance and mean of $\tilde{z}^{[l]}$.

9. After training a neural network with Batch Norm, at test time, to evaluate the neural network on a new example you should:

1 / 1 point

- ☒ Perform the needed normalizations, use μ and σ^2 estimated using an exponentially weighted average across mini-batches seen during training.
- ☐ Skip the step where you normalize using μ and σ^2 since a single test example cannot be normalized.
- ☐ If you implemented Batch Norm on mini-batches of (say) 256 examples, then to evaluate on one test example, duplicate that example 256 times so that you're working with a mini-batch the same size as during training.
- ☐ Use the most recent mini-batch's value of μ and σ^2 to perform the needed normalizations.

✔ Nice work

10. Which of these statements about deep learning programming frameworks are true? (Check all that apply)

0.7 / 1 point

- ☐ A programming framework allows you to code up deep learning algorithms with typically fewer lines of code than a lower-level language such as Python.
- ☐ Deep learning programming frameworks require cloud-based machines to run.
- ☒ Even if a project is currently open source, good governance of the project helps ensure that it remains open even in the long term, rather than become closed or modified to benefit only one company.

✔ Nice work

✘ You didn't select all the correct answers